ALM-1

① JRE:-

Definition:- JRE stands for Java Run time Environment

→ It provides the environment required to run java applications.

Roles of JRE:-

→ It runs java applications.

→ It contains JVM.

→ It loads required classes.

JDK:-

Definition:- JDK stands for Java development Key. It is a complete software package for java development.

Roles of JDK:-

→ It is used to develop java applications.
→ It is used to debug the java code
→ It is used to execute java programs.

② Diff b/w JVM, JRE and JDK.

JVM:-

1. It's full form is Java virtual Machine.
2. It's purpose is Runs Java bytecode.
3. It contains JVM only.
4. It can't compile code.

JRE:-

1. It's full form is Java Runtime Environment.
2. It provides environment to run java programs.
3. It contains JVM + core Java libraries.
4. It can't compile code.

JDK:-

1. It's full form is Java Development Kit
2. It provides tools to develop, compile and run java programs.
3. It contains JRE + development tools.
4. It can compile code.

③ Compilation and Execution flow diagram:-

Java Source program
↓
Java Compiler
↓
Bytecode
↓
Class loader
↓
Byte code
↓
Execution engine
↓
Machine code
↓
CPU executes → Output displayed

④ Importance of JVM, JDK, JRE.

JVM:-

→ It executes Java Byte code.
→ It will provides automatic memory management.
→ It will perform garbage collection.
→ It manages exceptions efficiently.

JDK:-

→ It include debugging tools for error detection.
→ It will generate documentation using javacode.
→ It packages applications into JAR files.
→ It provides all tools required for java development.

JRE:-

→ It provides the run time environment.
→ It includes essential java libraries.
→ It supports class loading and execution.
→ It will simplifies application development.